

BITEBOUND



Simon Völkl, Johannes Schieder
Mobile and Ubiquitous Computing | OTH Amberg-Weiden

Repository: github.com/s-voelkl/BiteBound

Motivation



Ziel: Hoch performante IoT **Architektur** für ESP32-Spiel

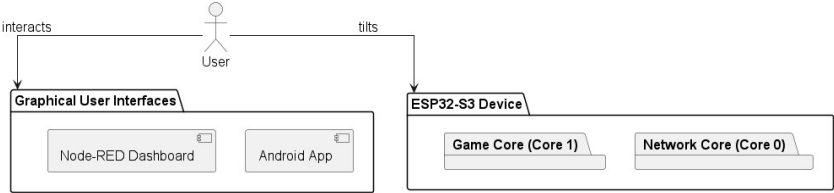


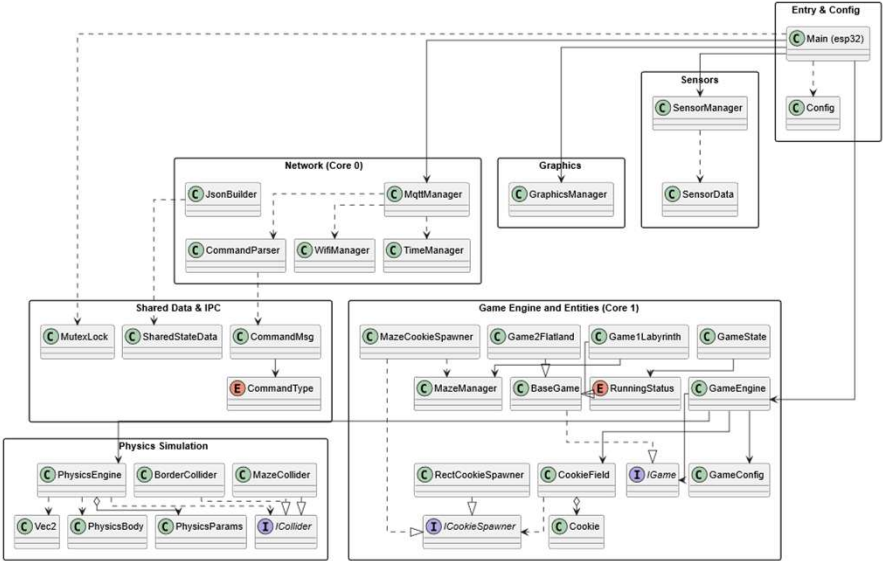
Risiko hoher **Latenz**

- Game & Physik Engine
- MQTT-Kommunikation
- Display



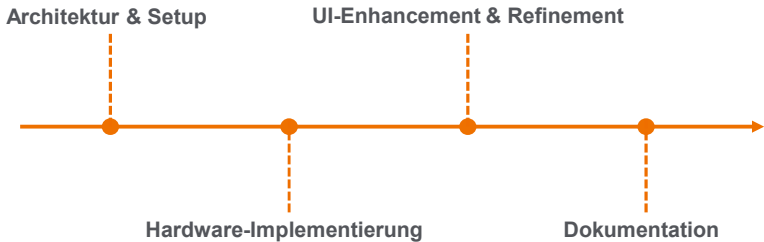
Prozessverteilung auf zwei Cores (dual-core)

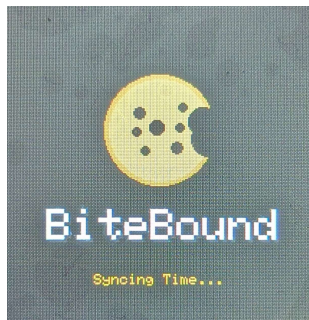




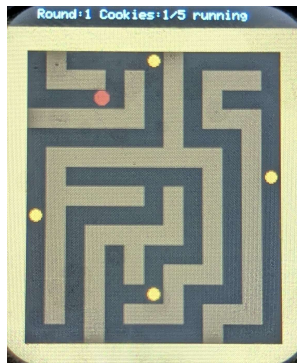
Hardware Architektur

4 Meilensteine



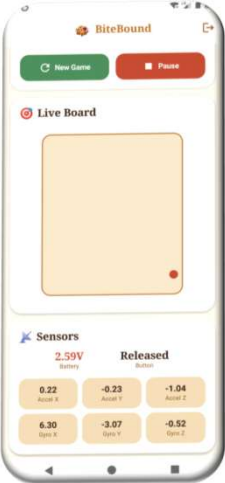
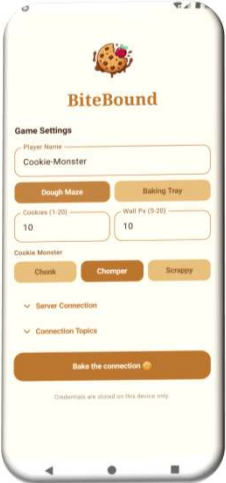


Loading Screen



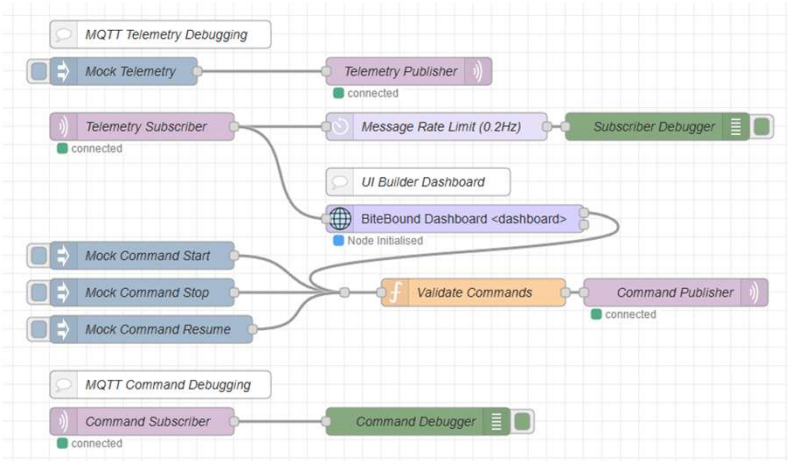
Loading Screen

Android



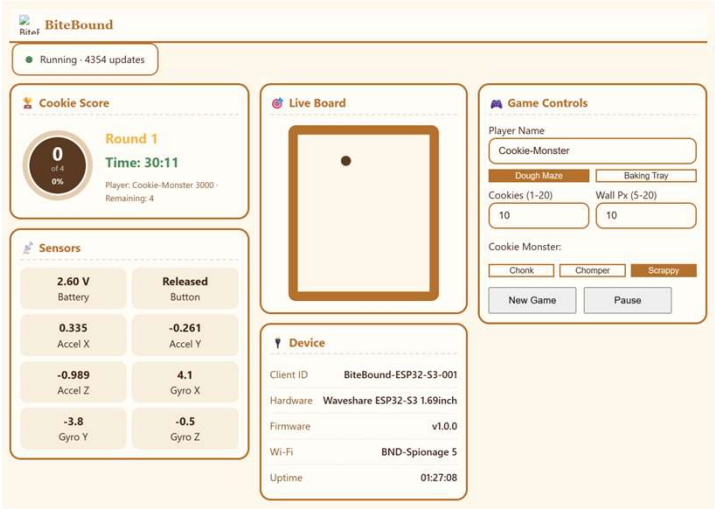
Node-RED

Workflow



Node-RED

UI Builder Dashboard



Code Qualität

Hardware & Frontend

Tests:

- Hardware (Arduino AUnit): **61 Unit Tests**
- Android Tests: **82%** Line-Coverage auf Data-Layer
- Mocks für MQTT-Nachrichten in Node-RED
- Kritisches Feedback durch Peer-Testing

GitHub Actions:

- CI/CD Pipelines
- Pull Request Requirement

 **All checks have passed**
2 successful checks

  CI / Compile ESP32 Sketch (production) (push) Successful in 5m

  CI / Compile ESP32 Sketch (tests, -DRUN_TESTS=1) (push) Successful in 5m

 **Merging is blocked**
At least 1 approving review is required by reviewers with write access.

Fazit & Lessons Learned



Summary, future work?

See `sec_summary.tex`

BITEBOUND



Simon Völkl, Johannes Schieder
Mobile and Ubiquitous Computing | OTH Amberg-Weiden

Repository: github.com/s-voelkl/BiteBound